

Lab 8 - Stiffening

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I. OBJECTIVES AND EXPERIMENTAL SETUP

The purpose of this lab was to investigate granular jamming structures and laminar jamming structures.

To make a granular jamming structure, we filled a balloon with rice, secured a tube to the opening. When the air gets sucked out of the balloon with a syringe while over a hook, the structure grasps onto the hook being able to lift objects. This is shown in figure 1.

The laminar jamming structures were fabricated by stuffing TPE pouches with different materials: layers of paper, rice, and oats. A tube was secured at the only opening, and a syringe sucked out the air to jam the structures. Testing included placing weights on the jammed structures while jammed and while not jammed, shown in figure 2 and 3.

II. RESULTS & ANALYSIS

Our experiments culminated in a deeper understanding of stiffening mechanisms in soft robotic applications. Not only did the compression material matter: so did the shape, size, and number of jamming grains which enabled the structure.

Through our experimental layered mechanism results, we found that the number of layers had a proportional relationship to stiffness. This would make sense, as more layers would cause more friction between layers. This means less slippage and bending. Based on the equations provided to us in the lab manual, this makes sense. We were given

$$\text{Stiffness} = \partial F \partial \delta \propto EI,$$

Where

$$I_{\text{prestip}} = \frac{bN^3h^3}{12}$$

$$I_{\text{unjammed}} = \frac{bNh^3}{12}$$

In combination with our results, these equations show a clear correlation between number of layers N and moment of inertia I , which would transversely relate N and stiffness proportionally. Stiffened and unstiffened moments are also shown, which makes clear a difference of N^2 . It is clear from these *why* the number of layers might affect the stiffness of a jamming structure.

Particle variation also affected the stiffness of jamming structures. Through our experimentation with different particles (rice and oats), we noticed that stiffer particles with more friction between one another resulted in stiffer structures. Small particles generally made for better grippers, as they could closely conform to objects. By using high-friction grains with varying shapes, high forces could be achieved throughout a structure. This resulted in stiffer and more compliant jammers. This analysis is shown in table 1 in the appendix.

The main difference in stiffness-generation between jammed and unjammed structures is the role of pressure. Noticeably, jamming requires negative pressure to compress layers or grains. This process is what strengthens the system. In unjammed actuators, high pressures are what led to stiffening. This was because there was no grain or layering to ‘hold’ a form and said negative pressure would just make a device more flimsy. However, positive pressure would lead to an outward force and an extension of all areas equally.

Layered jamming seems to be best in situations which require high normal forces. Resultant of the equations listed earlier, they are capable of generating large resistive forces. The drawback of this method is that it is only capable of this jamming in one direction (perpendicular to the layers). Granular jamming works much better in situations which require adaptation to dynamic structures. Gripping and locomotion on unusual surfaces could be particularly successful use cases for grain-based stiffening devices.

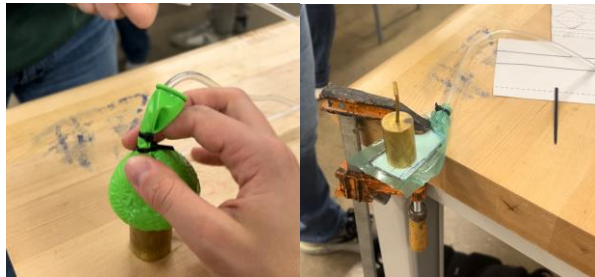


Figure 1: granular jamming structure (rice) Figure 2: laminar jamming structure (paper)



Figure 3: laminar jamming structures (rice and oats)

III. ASSUMPTIONS AND APPROXIMATIONS

One assumption made is that the fabrication created equally distributed structures and everything about the structures is uniform. Due to human error in fabrication and the irregularity of rice and oats, the structures are not necessarily uniform. It is also assumed that the cross section of the laminar structures is a rectangle. The laminar structures are also approximated as cantilever beams for analysis.

IV. CONCLUSION

This lab demonstrated the qualities of granular jamming structures and laminar jamming structures for grasping and jamming. These structures would be useful as soft robotic graspers, morphing structures for easy shape morphability, and impact absorption. Both are extremely versatile.

APPENDIX

Structure	Elastic Modulus	Deflection
Laminar Jamming Structure (paper)	13067 Pa	0.11 mm
Laminar Jamming Structure (rice)	338000 Pa	.052 mm
Laminar Jamming Structure (oats)	352000 Pa	.09 mm

Table 1: Analysis of Laminar Jamming Structures